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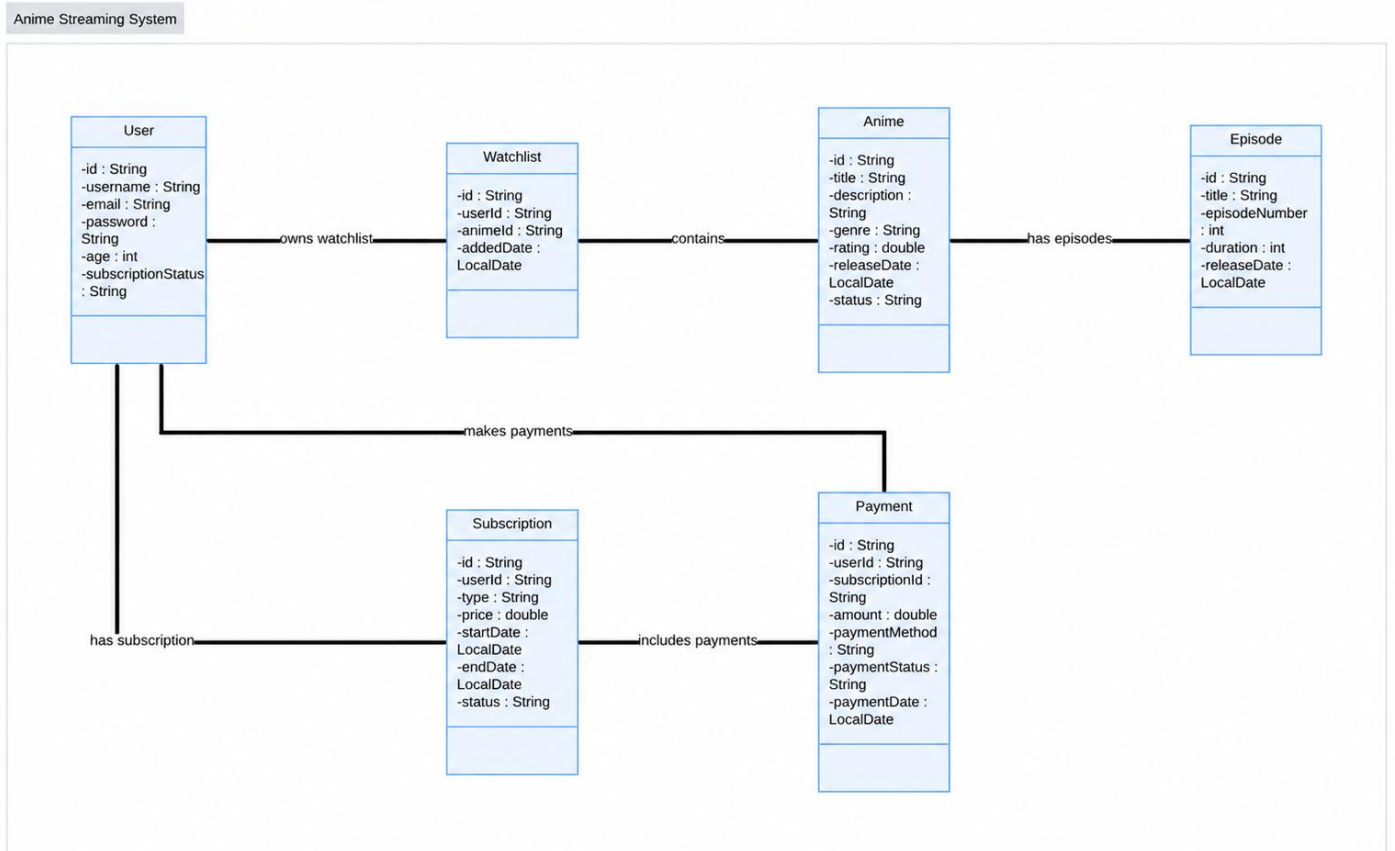
Overview of Crunchy roll system:

Crunchyroll is an online streaming platform mainly focused on anime content. The platform allows users to create accounts, browse and search for anime, view information about different shows, and watch available episodes.

Users can also manage their watchlist and keep track of the anime they are interested in. Crunchyroll provides different subscription plans, where subscribed users can access additional content and features depending on their plan.

The system includes information about users, anime, episodes, subscriptions, and user watchlists. These main components will be analyzed to identify the required models, variables, and validation rules.

UML Diagram:



Pic 1

Model User

Description:

Represents a registered user of the Crunchyroll platform and stores the user's basic account and subscription information.

The following table presents the User model attributes, their data types, and the validation requirements for each attribute.

Attribute	Data Type	Requirements
id	String	1. Cannot be null or empty 2. length must be at least 3 characters
username	String	1. Cannot be null or empty 2. length must be at least 3 characters 3. must contain letters only
email	String	1. Cannot be null or empty 2. must be a valid email address.
password	String	1. Cannot be null or empty 2. must be at least 8 characters 3. must contain at least one uppercase letter, one lowercase letter, one number, and one special character
age	int	1. Cannot be null 2. must be at least 18 years old
subscriptionType	String	1. Cannot be null or empty 2. must be either Fan, Mega Fan, or Ultimate Fan

Table 1

User Model Validation Annotations:

```
public class User { no usages

    @NotEmpty(message = "id cannot be empty") no usages
    @Size(min = 3, message = "id must be at least 3 characters")
    private String id;

    @NotEmpty(message = "username cannot be empty") no usages
    @Size(min = 3, message = "username must be at least 3 characters")
    @Pattern(regexp = "[a-zA-Z ]+$", message = "Username must contain letters only")
    private String username;

    @NotEmpty(message = "email cannot be empty") no usages
    @Email(message = "email must be valid")
    private String email;

    @NotEmpty(message = "password cannot be empty") no usages
    @Size(min = 8, message = "password must be at least 8 characters")
    @Pattern(regexp = "^(?=.*[a-z])(?=.*[A-Z])(?=.*\\d)(?=.*[a-zA-Z0-9!@#$%^&*~`-_|{}~:;\"' ,./\\?])[a-zA-Z0-9!@#$%^&*~`-_|{}~:;\"' ,./\\?]+$", message = "password must contain uppercase, lowercase, number and special character")
    private String password;

    @NotNull(message = "age cannot be null") no usages
    @Min(value = 18, message = "age must be at least 18")
    private int age;

    @NotEmpty(message = "subscription type cannot be empty") no usages
    @Pattern(regexp = "Fan|Mega Fan|Ultimate Fan", message = "subscription type must be Fan, Mega Fan, or Ultimate Fan")
    private String subscriptionType;
}
```

pic 2

Model Anime

Description:

Represents an anime available on the platform and stores its main information, such as title, description, genre, rating, release date, and status.

The following table presents the Anime model attributes, their data types, and the validation requirements for each attribute.

Attribute	Data Type	Requirements
id	String	1. Cannot be null or empty 2. length must be at least 3 characters
title	String	1. Cannot be null or empty 2. length must be at least 2 characters
description	String	1. Cannot be null or empty 2. Length must be at least 10 characters
genre	String	1. Cannot be null or empty 2. Must contain letters only
rating	double	1. Cannot be null 2. Must be between 0 and 5
releaseDate	LocalDate	1. Cannot be null or empty 2. Cannot be a future date
status	String	1. Cannot be null or empty 2. Must be either Ongoing or Completed

Table 2

Anime Model Validation Annotations:

```
public class Anime { no usages
    @NotEmpty(message = "id cannot be empty") no usages
    @Size(min = 3, message = "id must be at least 3 characters")
    private String id;

    @NotEmpty(message = "title cannot be empty") no usages
    @Size(min = 2, message = "title must be at least 2 characters")
    private String title;

    @NotEmpty(message = "description cannot be empty") no usages
    @Size(min = 10, message = "description must be at least 10 characters")
    private String description;

    @NotEmpty(message = "genre cannot be empty") no usages
    @Pattern(regexp = "[a-zA-Z ]+$", message = "genre must contain letters only")
    private String genre;

    @NotNull(message = "rating cannot be null") no usages
    @DecimalMin(value = "0.0", message = "rating must be at least 0")
    @DecimalMax(value = "5.0", message = "rating cannot be more than 5")
    private double rating;

    @NotNull(message = "release date cannot be null") no usages
    @PastOrPresent(message = "release date cannot be in the future")
    private LocalDate releaseDate;

    @NotEmpty(message = "status cannot be empty") no usages
    @Pattern(regexp = "Ongoing|Completed", message = "status must be either Ongoing or Completed")
    private String status;
}
```

Pic 2

Model Episode:

Description:

Represents an individual episode of an anime available on the platform and stores information such as the episode title, episode number, duration, and release date.

The following table presents the Episode model attributes, their data types, and the validation requirements for each attribute.

Attribute	Data Type	Requirements
id	String	1. Cannot be null or empty 2. length must be at least 3 characters
title	String	1. Cannot be null or empty 2. length must be at least 2 characters
episodeNumber	int	1. Cannot be null or empty 2. Must be a positive number
duration	int	1. Cannot be null or empty 2. Must be a positive number
releaseDate	LocalDate	1. Cannot be null or empty 2. Cannot be a future date

Table 3

Episode Model Validation Annotations:

```
public class Episode {  
    @NotEmpty(message = "id cannot be empty")  
    @Size(min = 3, message = "id must be at least 3 characters")  
    private String id;  
  
    @NotEmpty(message = "title cannot be empty")  
    @Size(min = 2, message = "title must be at least 2 characters")  
    private String title;  
  
    @NotNull(message = "episode number cannot be null")  
    @Positive(message = "episode number must be a positive number")  
    private int episodeNumber;  
  
    @NotNull(message = "duration cannot be null")  
    @Positive(message = "duration must be a positive number")  
    private int duration;  
  
    @NotNull(message = "release date cannot be null")  
    @PastOrPresent(message = "release date cannot be in the future")  
    private LocalDate releaseDate;  
}
```

Pic 3

Model Watchlist:

Description:

Represents a user's watchlist and stores the anime added by the user for later viewing.

The following table presents the Watchlist model attributes, their data types, and the validation requirements for each attribute.

Attribute	Data Type	Requirements
id	String	1. Cannot be null or empty 2. length must be at least 3 characters
userId	String	1. Cannot be null or empty 2. length must be at least 3 characters
animeId	String	1. Cannot be null or empty 2. length must be at least 3 characters
addedDate	LocalDate	1. Cannot be null or empty 2. Cannot be a future date

Table 4

Watchlist Model Validation Annotations:

```
public class Watchlist {  
  
    @NotEmpty(message = "id cannot be empty")  
    @Size(min = 3, message = "id must be at least 3 characters")  
    private String id;  
  
    @NotEmpty(message = "user id cannot be empty")  
    @Size(min = 3, message = "user id must be at least 3 characters")  
    private String userId;  
  
    @NotEmpty(message = "anime id cannot be empty")  
    @Size(min = 3, message = "anime id must be at least 3 characters")  
    private String animeId;  
  
    @NotNull(message = "added date cannot be null")  
    @PastOrPresent(message = "added date cannot be in the future")  
    private LocalDate addedDate;  
}
```

Pic 4

Model Subscription:

Description:

Represents a user's subscription to the platform and stores information about the selected subscription type, price, subscription period, and current status.

The following table presents the Subscription model attributes, their data types, and the validation requirements for each attribute.

Attribute	Data Type	Requirements
id	String	1. Cannot be null or empty 2. length must be at least 3 characters
userId	String	1. Cannot be null or empty 2. length must be at least 3 characters
type	String	1. Cannot be null or empty 2. Must be either Fan, Mega Fan, or Ultimate Fan
price	double	1. Cannot be null or empty 2. Must be a positive number
startDate	LocalDate	1. Cannot be null or empty 2. Cannot be a future date
endDate	LocalDate	1. Cannot be null or empty 2. Must be a future date
status	String	1. Cannot be null or empty 2. Must be either Active, Expired, or Canceled

Table 5

Subscription Model Validation Annotations:

```
public class Subscription { no usages

    @NotEmpty(message = "id cannot be empty") no usages
    @Size(min = 3, message = "id must be at least 3 characters")
    private String id;

    @NotEmpty(message = "user id cannot be empty") no usages
    @Size(min = 3, message = "user id must be at least 3 characters")
    private String userId;

    @NotEmpty(message = "subscription type cannot be empty") no usages
    @Pattern(regexp = "Fan|Mega Fan|Ultimate Fan", message = "type must be Fan, Mega Fan, or Ultimate Fan")
    private String type;

    @NotNull(message = "price cannot be null") no usages
    @Positive(message = "price must be a positive number")
    private double price;

    @NotNull(message = "start date cannot be null") no usages
    @PastOrPresent(message = "start date cannot be in the future")
    private LocalDate startDate;

    @NotNull(message = "end date cannot be null") no usages
    @Future(message = "end date must be in the future")
    private LocalDate endDate;

    @NotEmpty(message = "status cannot be empty") no usages
    @Pattern(regexp = "Active|Expired|Canceled", message = "status must be Active, Expired, or Canceled")
    private String status;
}
```

Pic 5

Model Payment:

Description:

Represents a payment made by a user for a subscription and stores information about the payment amount, payment method, status, and payment date.

The following table presents the Payment model attributes, their data types, and the validation requirements for each attribute.

Attribute	Data Type	Requirements
id	String	1. Cannot be null or empty 2. length must be at least 3 characters
userId	String	1. Cannot be null or empty 2. length must be at least 3 characters
subscriptionId	String	1. Cannot be null or empty 2. Length must be at least 3 characters
amount	double	1. Cannot be null or empty 2. Must be a positive number
paymentMethod	String	1. Cannot be null or empty 2. Must be either Mada, STC Pay, or Credit Card
paymentStatus	String	1. Cannot be null or empty 2. Must be either Pending, Completed, or Failed
paymentDate	LocalDate	1. Cannot be null or empty 2. Cannot be a future date

Table 6

Payment Model Validation Annotations:

```
public class Payment { no usages

    @NotEmpty(message = "id cannot be empty") no usages
    @Size(min = 3, message = "id must be at least 3 characters")
    private String id;

    @NotEmpty(message = "user id cannot be empty") no usages
    @Size(min = 3, message = "user id must be at least 3 characters")
    private String userId;

    @NotEmpty(message = "subscription id cannot be empty") no usages
    @Size(min = 3, message = "subscription id must be at least 3 characters")
    private String subscriptionId;

    @NotNull(message = "amount cannot be null") no usages
    @Positive(message = "amount must be a positive number")
    private double amount;

    @NotEmpty(message = "payment method cannot be empty") no usages
    @Pattern(regexp = "Mada|STC Pay|Credit Card", message = "payment method must be Mada, STC Pay, or Credit Card")
    private String paymentMethod;

    @NotEmpty(message = "payment status cannot be empty") no usages
    @Pattern(regexp = "Pending|Completed|Failed", message = "payment status must be Pending, Completed, or Failed")
    private String paymentStatus;

    @NotNull(message = "payment date cannot be null") no usages
    @PastOrPresent(message = "payment date cannot be in the future")
    private LocalDate paymentDate;
}
```

Pic 6